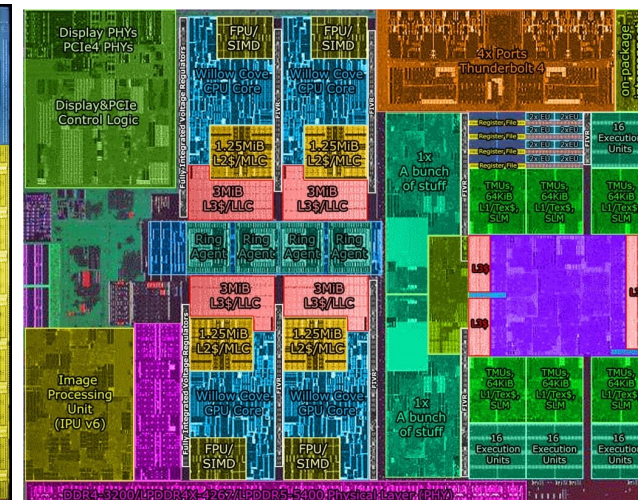
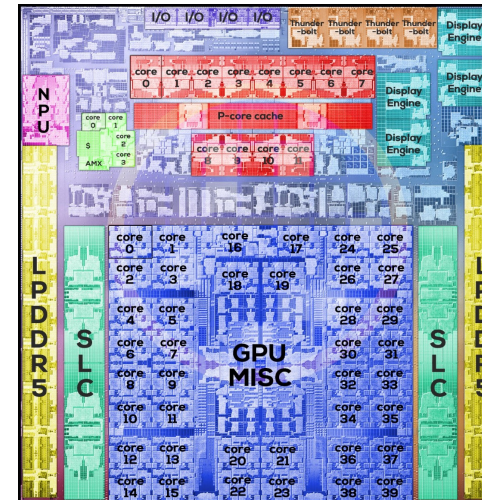
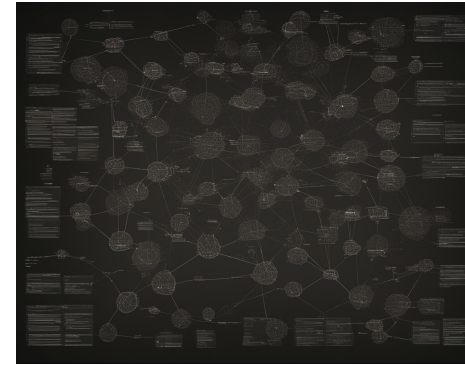


Securing Large Monolithic Systems: Challenges and Opportunities

Saket Upadhyay, Arnabjyoti Kalita, Ashish Venkat

The Landscape of Modern Computing

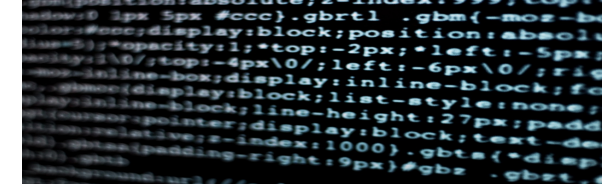
Software (rapidly evolving, abstract, diverse)



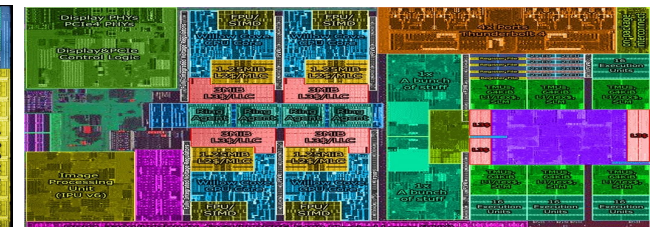
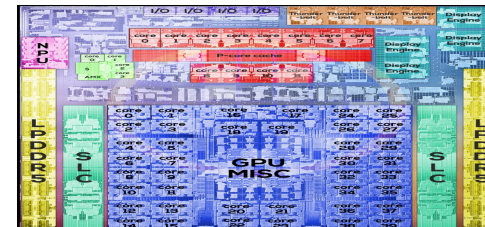
Hardware (highly heterogeneous, complex, obscure)

The Landscape of Modern Computing

Software (rapidly evolving, abstract, diverse)



Widening Hardware-Software Gap



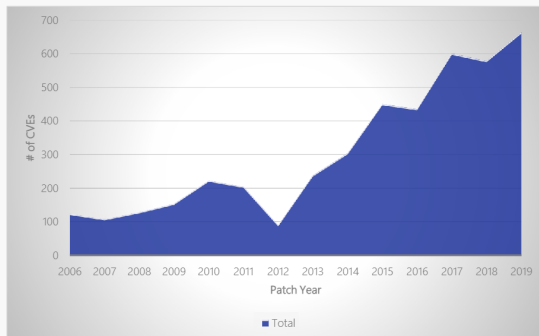
4 Hardware (highly heterogeneous, complex, obscure)

The Widening Hardware-Software Gap

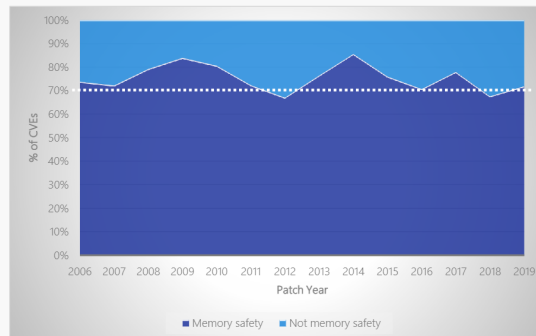
Security Implications – Explosion of vulnerabilities

It is too easy to do the unsafe thing today

For systems software[1] at Microsoft, memory safety is the most common vulnerability class



Vulnerabilities reported & fixed per year is increasing



~70% of vulnerabilities are memory safety year over year

Memory safety is an industry challenge [2,3,4,5]

~66% of iOS 12 vulnerabilities

~72% of macOS 10.14 vulnerabilities

~60% of high severity vulnerabilities in Chrome

~90% of Android vulnerabilities

* Matt Miller, Microsoft. WOOT 2019 Presentation

Legacy C/C++ code is a nuclear waste nightmare that will make you WannaCry

June 11, 2017 ∞

* Trevor Jim, AT&T Labs. Blog Article.

LILY HAY NEWMAN

MATT BURGESS

ANDY GREENBERG

SECURITY JUL 19, 2024 10:46 AM

How One Bad CrowdStrike Update Crashed the World's Computers

A defect
reboot c
it. Here's

The Urgent Need for Memory Safety in Software Products

Released: September 20, 2023

Revised: December 06, 2023

Bob Lord, Senior Technical Advisor



CES, ORGANIZATIONS AND CYBER SAFETY

THE WALL STREET JOURNAL.

Tech Giants Race to Address Chip Flaws With a Potentially Vast Impact

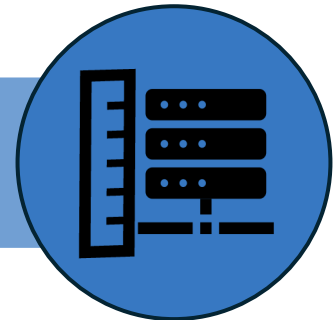
Spectre and Meltdown, long-time design bugs, could make many devices vulnerable to hack

Necessitates systems that ...



effectively limit damage even upon initial penetration

support secure & scalable sys. resource sharing

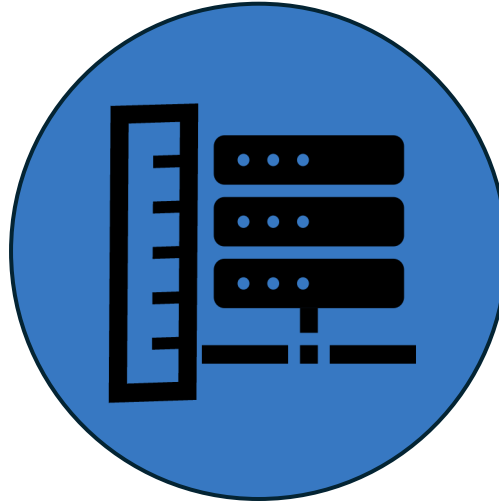


can be enabled on-demand and in the field

How we have been tackling this!



Secure Compartmentalization



Secure Virtualization



Field-Upgradable Defenses

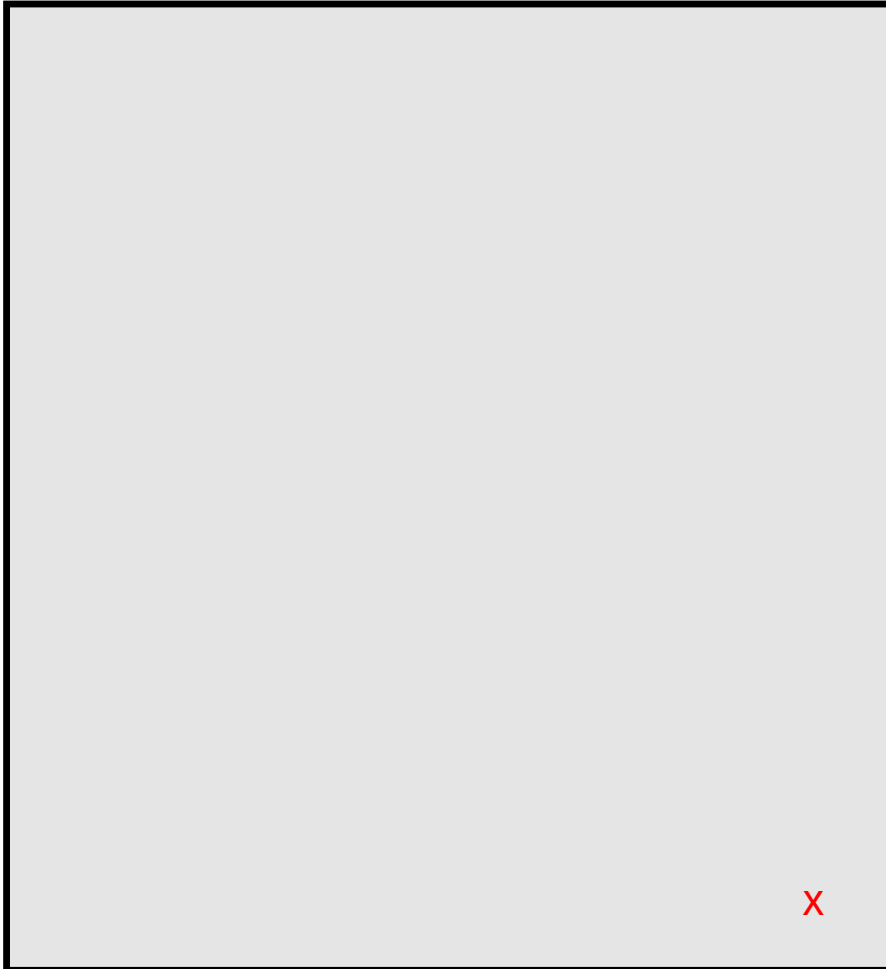
Secure Compartmentalization

Breaking monolithicity by defining fine-grain execution policies.

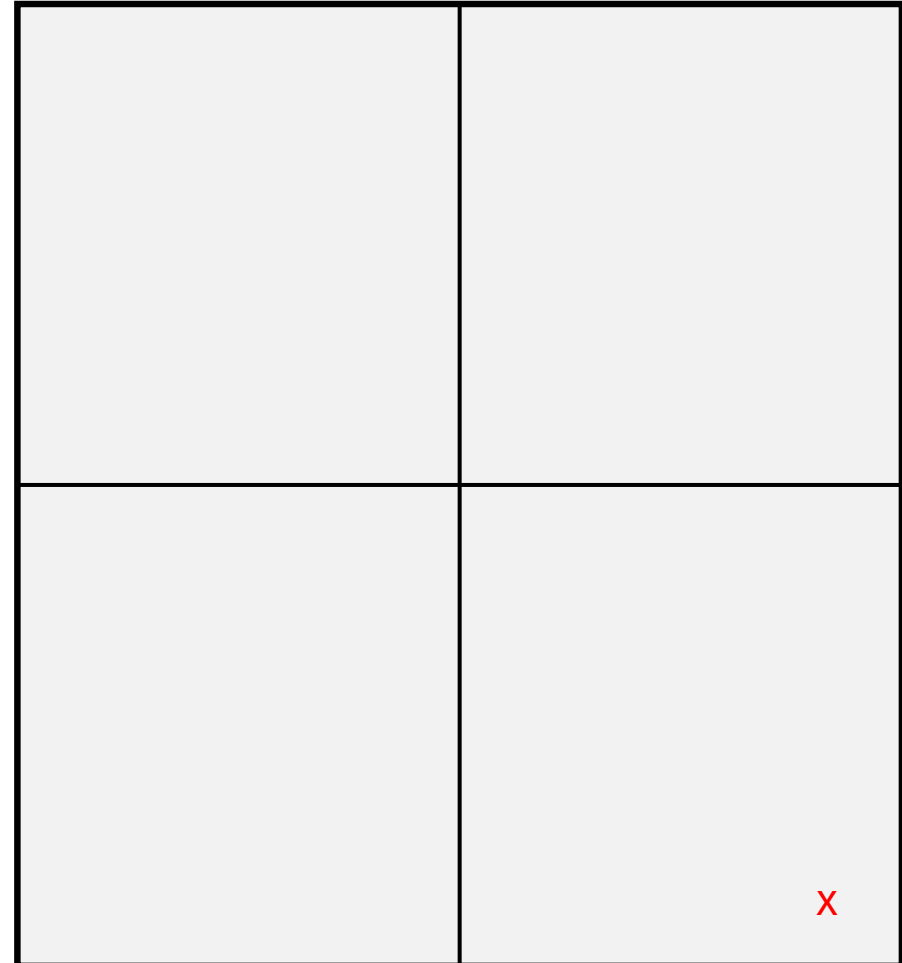


The idea of compartmentalization

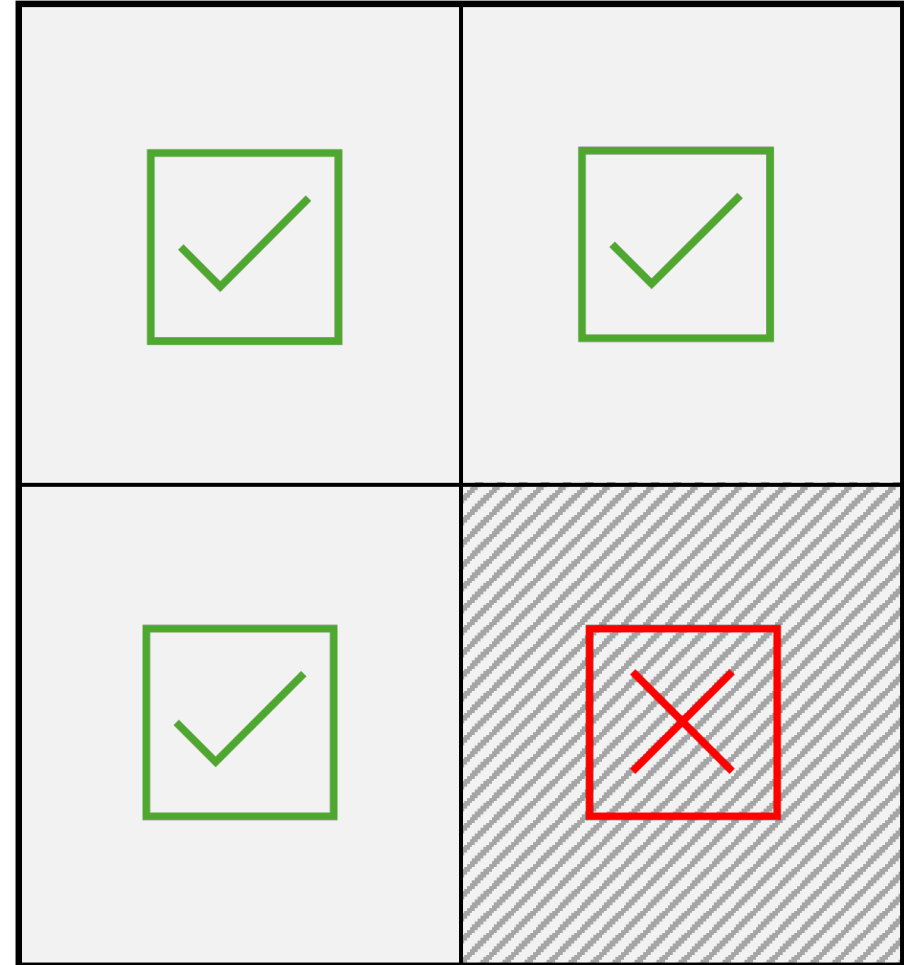
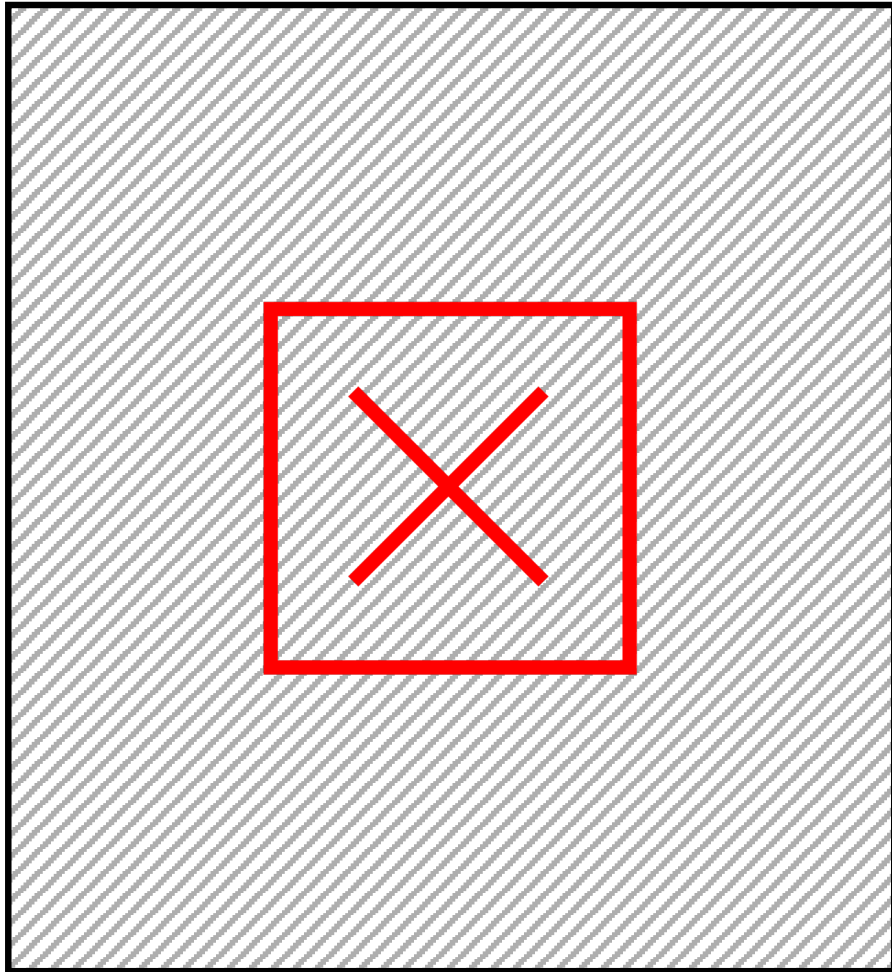
single panel window



multi-panel window



The idea of compartmentalization



The Conundrum of Large Monolithic Software

[illegible]

[illegible]

Abstract model: Core Terms

Principal: An active entity playing a specific role.
(Think of function or group of functions)

Compartment: Collection of data with a specific purpose. (Think of structured data- struct, class, etc.)

Access rules:

- For each compartment and each principal, a matrix specifying which operations are legitimate
- Collectively forms a policy restricting flow of data between compartments and integrity of data within each compartment

Load	$\mathcal{C}_1$	$\mathcal{C}_2$	$\mathcal{C}_3$	$\mathcal{C}_4$	$\mathcal{C}_5$	...	$\mathcal{C}_n$
Principal-1	Y	Y	trap	trap	trap		trap
Principal-2	trap	Y	Y	Y	trap		trap
Principal-3	Y	trap	trap	trap	Y		trap
⋮							
Principal-n	trap	trap	Y	trap	trap		Y



Effective Compartment Requirements

- Refactoring of large monolithic systems into appropriate subject domains.
- Derivation of fine-grained access control policies for a given subject domain-object domain pair.
- Mechanisms responsible for enforcing the derived access control policies.

Challenges

COMPILATION

- Novel strategies for automatic code refactoring: reduce manual effort in compartmentalizing large codebases.
- Effective expense modeling to make tradeoffs between performance and security.
- Soundness and precision of static analysis schemes and coverage of dynamic analysis.
- Scalability to support large codebases and complex privilege lattices.

SOFTWARE-HARDWARE HANDOFF

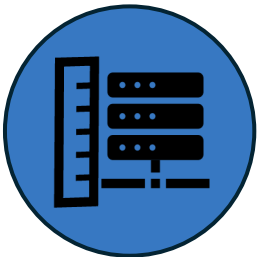
- Novel interchange formats for size & time efficient communication of metadata to the hardware for secure and effective policy enforcement.
- Hardware-software co-optimization strategies-
 - Software-based policy derivation schemes tuned to underlying hardware-based enforcement.
 - Performance hints (e.g., prefetch hints, subject/context-switching gate call prediction hints, etc.)
 - Performant execution of a compartmentalized workload.

HARDWARE

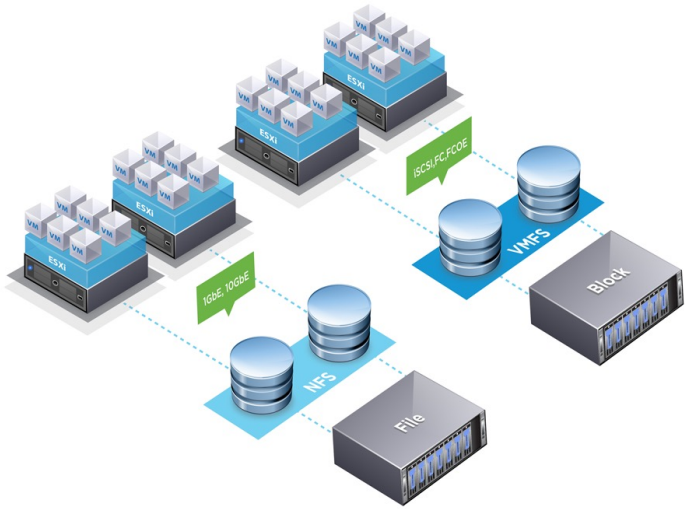
Novel strategies to compartmentalize large monolithic hardware architectures that stack layers of intricate microarchitectural functionalities.

Secure Virtualization

by fine-grain virtual resource access control.



What is virtualized?



Hyper-V Network Virtualization

Article • 10/27/2021 • 9 contributors



Microsoft
Hyper-V

Memory Virtualization with vSphere



Physical storage architectures for Hyper-V

Article • 06/24/2024 • 2 contributors



openstack®

CPU Virtualization in vSphere

What is virtualized?

- CPUs
- Memory
- Storage Devices
- NIC
- GPU
- Storage Controllers
- Peripherals
- Fiber Channel Interfaces
- iSCSI Interfaces
- Network Appliances
- Motherboards
- TPM
- Serial Ports
- Parallel Ports

A lot more in academia...

Secure virtualization for cloud computing

[F Lombardi](#), [R Di Pietro](#) - Journal of network and computer applications, 2011 - Elsevier

... **security** issues that affect both the cloud provider and the cloud user. In this paper, we show how **virtualization** can increase the **security** ... increased **security** to cloud **resources**. ACPS can ...

☆ Save 📄 Cite Cited by 611 Related articles All 7 versions

Virtualization: Issues, security threats, and solutions

[M Pearce](#), [S Zeadally](#), [R Hunt](#) - ACM Computing Surveys (CSUR), 2013 - dl.acm.org

... **security** implications that can result from weak **virtualization** ... We discuss **security** implications resulting from the **virtualization** ... This is due to the **resource** control property of **virtualization**, ...

☆ Save 📄 Cite Cited by 333 Related articles All 8 versions 📄

Security issues with virtualization in cloud computing

[V Kumar](#), [RS Rathore](#) - 2018 International Conference on ..., 2018 - ieeeexplore.ieee.org

... **resource** pooling with the help of **virtualization**. For **secure resource** pooling, we need a **secure** ... Hypervisors help in **secure** isolation of multiple virtual machines running on single ...

☆ Save 📄 Cite Cited by 55 Related articles

A first step toward network security virtualization: From concept to prototype

[S Shin](#), [H Wang](#), [G Gu](#) - ... on Information Forensics and Security, 2015 - ieeeexplore.ieee.org

... new concept of Network **Security Virtualization** (NSV), which virtualizes **security resources**/functions to ... only when necessary, we can efficiently manage network **security resource** usage. ...

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NOVA: A microhypervisor-based secure virtualization architecture

[U Steinberg](#), [B Kauer](#) - Proceedings of the 5th European conference on ..., 2010 - dl.acm.org

... the platform, the **virtualization** layer facilitates sharing of **resources** and decouples the guest ... interactions between virtual machines and the physical **resources** of the host system. Each ...

☆ Save 📄 Cite Cited by 542 Related articles All 14 versions

Virtualization security for cloud computing service

[S Luo](#), [Z Lin](#), [X Chen](#), [Z Yang](#) - ... Conference on Cloud and ..., 2011 - ieeeexplore.ieee.org

... **resource** layer contains the required hardware **resources** of the ... out some **security** vulnerabilities in **virtualization security** and ... Then we propose a **virtualization security** framework ...

☆ Save 📄 Cite Cited by 167 Related articles All 4 versions

[PDF] Challenges in resource allocation in network virtualization

[A Haider](#), [R Potter](#), [A Nakao](#) - 20th ITC specialist seminar, 2009 - itcspecialistseminar.com

... the **virtualization** of physical **resources**. Hence, a pool of virtual **resources** can be created by a **virtualization** layer that implements the abstraction of physical **resources** available at SN. ...

☆ Save 📄 Cite Cited by 187 Related articles All 3 versions 📄

Multi-granularity resource virtualization and sharing strategies in cloud manufacturing

[N Liu](#), [X Li](#), [W Shen](#) - Journal of Network and Computer Applications, 2014 - Elsevier

... It encapsulates physical **resources** into cloud services ... **resource virtualization** and sharing strategies for bridging the gap between complex manufacturing tasks and underlying **resources**...

☆ Save 📄 Cite Cited by 99 Related articles All 4 versions

Wireless resource virtualization with device-to-device communication underlying LTE network

[A Moubayed](#), [A Shami](#), [H Lutfiyya](#) - IEEE Transactions on ..., 2015 - ieeeexplore.ieee.org

... **Virtualization** allows for more efficient utilization of the spectrum, reduces capital ... In this paper, the problem of wireless **resource virtualization** with D2D communication underlying the ...

☆ Save 📄 Cite Cited by 87 Related articles All 6 versions

Virtualization and cloud computing

[Y Xing](#), [Y Zhan](#) - Future Wireless Networks and Information Systems ..., 2012 - Springer

... Virtual **resources** layer will integrate and manage the **resources** what have been ... **Resources Virtualization** layer processes the **virtualization** of all kinds of the computing **resources**, and it ...

☆ Save 📄 Cite Cited by 172 Related articles All 4 versions

An SLA-based resource virtualization approach for on-demand service provision

[A Kertesz](#), [G Kecskemeti](#), [I Brandic](#) - ... workshop on Virtualization ..., 2009 - dl.acm.org

... consider either **virtualization** approaches [12... **resource** virtualizations [27] [6]. In this paper we propose a novel holistic architecture considering **resource** provision using the **virtualization** ...

☆ Save 📄 Cite Cited by 99 Related articles All 12 versions

A summary of virtualization techniques

[F Rodríguez-Haro](#), [F Freitag](#), [L Navarro](#) - ... Procedia ..., 2012 - Elsevier

... **virtualization** techniques from the OS view. First, we discuss two techniques that executes modified guest OSs: operating system-level **virtualization** and para-**virtualization**. ... of **resource** ...

☆ Save 📄 Cite Cited by 105 Related articles All 8 versions

Saket Upadhyay | ESWeek '24

Security issues and challenges for virtualization technologies

[F Sierra-Arriaga](#), [R Branco](#), [B Lee](#) - ACM Computing Surveys (CSUR), 2020 - dl.acm.org

... to share platform **resources**, hypervisors are often expected to provide **security** boundaries between ... of **resources**) add to complexity of analyzing **security** for virtualized environments. ...

☆ Save 📄 Cite Cited by 42 Related articles All 2 versions

A review of edge computing: Features and resource virtualization

[Y Mansouri](#), [MA Babar](#) - Journal of Parallel and Distributed Computing, 2021 - Elsevier

... SCC is a double-edged sword cloud-based paradigm since it imposes several overheads such as management and orchestration of **resources** and requires a **security** mechanism to ...

☆ Save 📄 Cite Cited by 151 Related articles

Secure service virtualization in IoT by dynamic service dependency verification

[H Ko](#), [J Jin](#), [SL Keoh](#) - IEEE Internet of Things Journal, 2016 - ieeeexplore.ieee.org

... it can retrieve the client's **resource** stored in the server. In OAuth, granting a third-party access to a client's **resource** is analogous to granting service **virtualization** in SDT. However, SDT ...

☆ Save 📄 Cite Cited by 24 Related articles All 6 versions

vNIDS: Towards elastic security with safe and efficient virtualization of network intrusion detection systems

[H Li](#), [H Hu](#), [G Gu](#), [GJ Ahn](#), [F Zhang](#) - ... and Communications Security, 2018 - dl.acm.org

... The safe **virtualization** requires that virtualized NIDSes do not miss any ... **virtualization** requires that virtualized NIDSes are provisioned optimally and consume minimum **resources**. ...

☆ Save 📄 Cite Cited by 61 Related articles All 9 versions

From virtualization security issues to cloud protection opportunities: An in-depth analysis of system virtualization models

[M Compastié](#), [R Badonnel](#), [Q Festor](#), [R He](#) - Computers & Security, 2020 - Elsevier

... As the cloud paradigm requires the decoupling of software **resources** from hardware **resources**, system **virtualization** has become one of the enablers of this paradigm. ...

☆ Save 📄 Cite Cited by 48 Related articles All 7 versions

Secure multi-cloud network virtualization

[M Alaluna](#), [E Vial](#), [N Neves](#), [FMV Ramos](#) - Computer Networks, 2019 - Elsevier

... data center context, scales poorly or does not consider the **security** of virtual **resources**. We propose a scalable heuristic that considers **security** as a first-class citizen and is specifically ...

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Embedded virtualization for the design of secure IoT applications

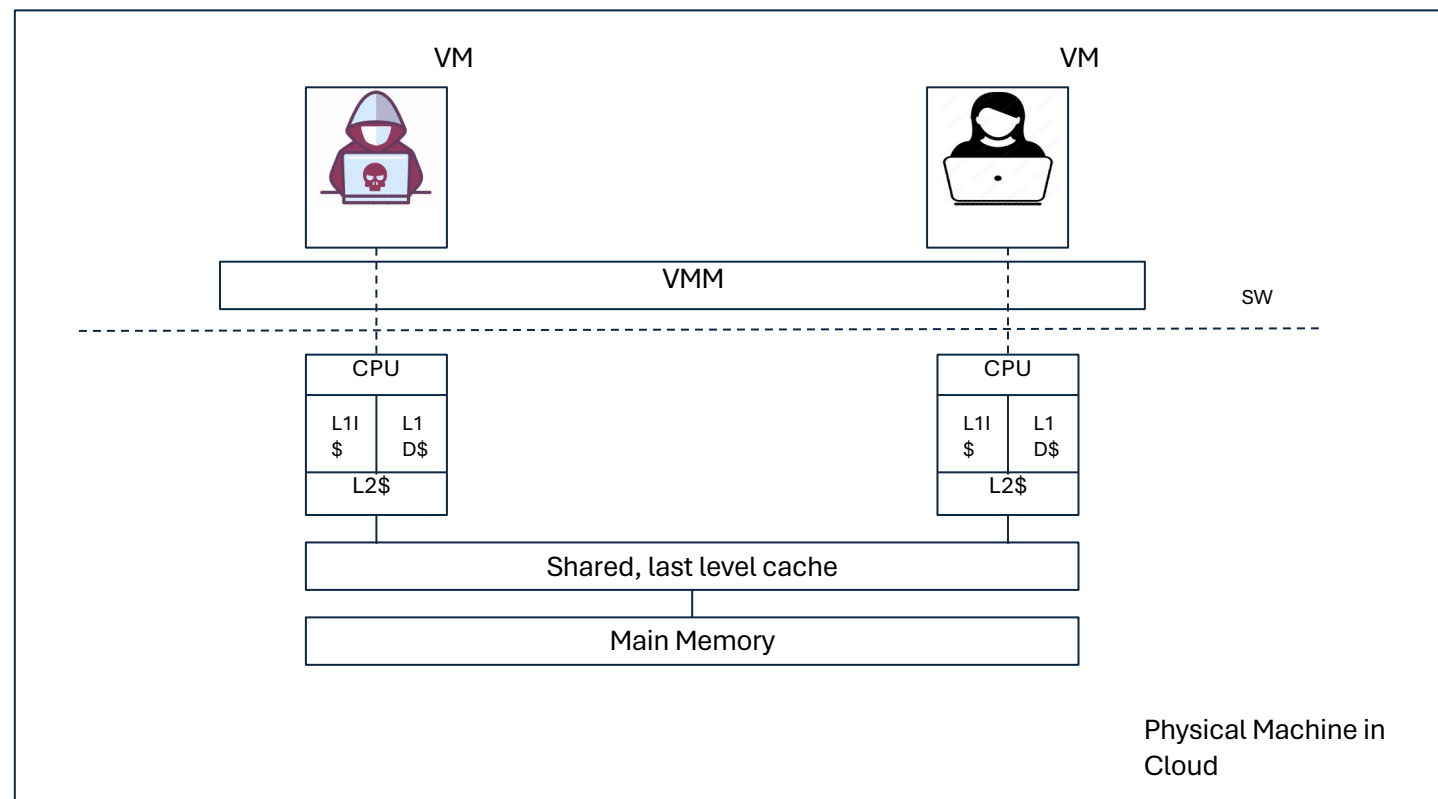
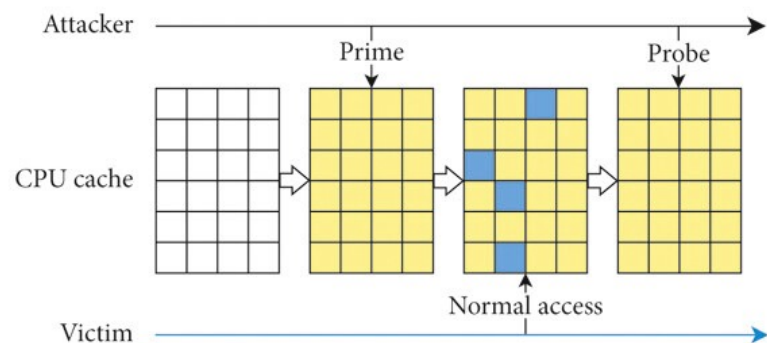
[C Moratelli](#), [S Johann](#), [M Neves](#), [F Hessel](#) - Proceedings of the 27th ..., 2016 - dl.acm.org

... a possible approach to build more **secure** IoT devices. Moreover, all advantages of general ... **virtualization** will be inherited by IoT devices. At first glance, **virtualization** on such **resource** ...

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Security Concerns

- Side-channel inference
- Crosstalk



Co-located attackers that compete for shared physical resources may covertly infer or influence a victim's behavior.

Approach

- Access must follow the principle of least privilege, granting only necessary resources.
- Operations require explicit verification of access rights before execution.
- Resource sharing and revocation should be explicit, voluntary, and limited to trusted domains.
- Maintain a minimum allocation guarantee and prevent exceeding maximum thresholds.

Ceviche: Capability Enhanced Virtualization of Caches

- A capability-based cache lookup by translating a given address-domain ID pair into a capability.
- Encodes the access rights and the allowed set of operations on the physical cache line that it grants access to.
- Constrains cache lookup based on a capability, can achieve fine-grained partitioning of the cache at the granularity of a cache line.
- Decouples the address from the physical location of a cache line, enables fully associative allocation and fine-grained cache partitioning.

Challenges

PERFORMANCE OVERHEAD

- Virtualizing fine-grained system-level resources adds an additional layer of indirection.
- Incurs resource lookup penalty for every access.
- We need novel and optimized lookup procedures where the performance cost of the lookup is minimized.

STORAGE OVERHEAD

- A substantial number of metadata encoding access rights to be tracked for per user per virtualized resource in the system that is either temporally or spatially shared.
- We need scalable novel data structures or hardware organizations that minimize the storage overheads.

Field-Upgradable Defenses

Fast, reliable, transparent deployment of defenses supporting legacy code.



State-of-the-Art Defenses

```
void foo (int key, int v)
{
    char c[12];
    if (key < 12)
        c[key] = v;
    ...
}
```

Bounds Checking

```
if (key[i] == 1)
    x = x^2 * m;
else
    dummy_mult;
    x = x^2
```

Constant-Time Execution

```
void foo ()
{
    ...
    canary_check;
    return;
}
```

Canary Checks

```
mov %rcx, 1000000c35841h
mov %rcx, 3BF43B1820E7ED7Dh
mov %rdx, 3BF53B182024B53Ch
xor %rcx, %rdx
mov qword ptr [rax+48h], %rcx
```

Constant Blinding

State-of-the-Art Defenses

✗ Recompilation / Binary Translation

✗ Significant Performance Overhead

✗ Visible to Attackers

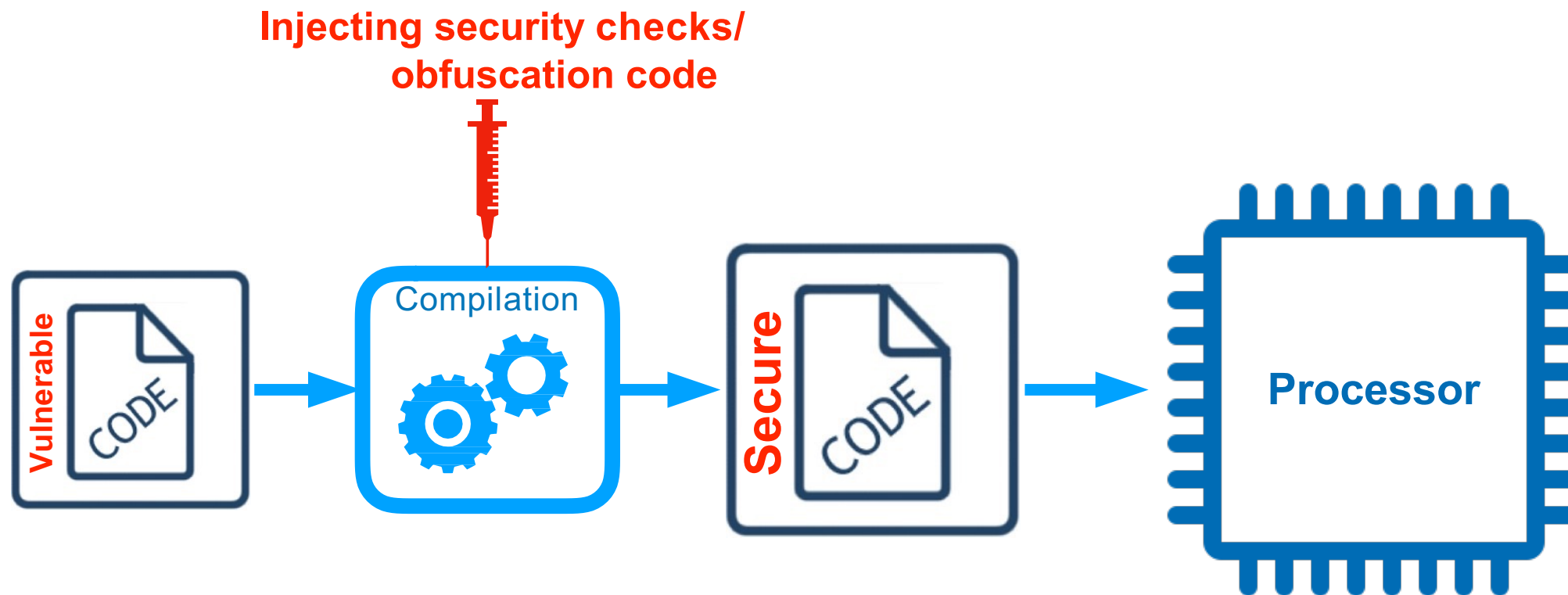
```
void foo (int key, int v)
{
    char c[12];
    if (key < 12)
        c[key] = v;
    ...
}
```

```
if (key[i] == 1)
    x = x^2 * m;
else
    x = x^2
```

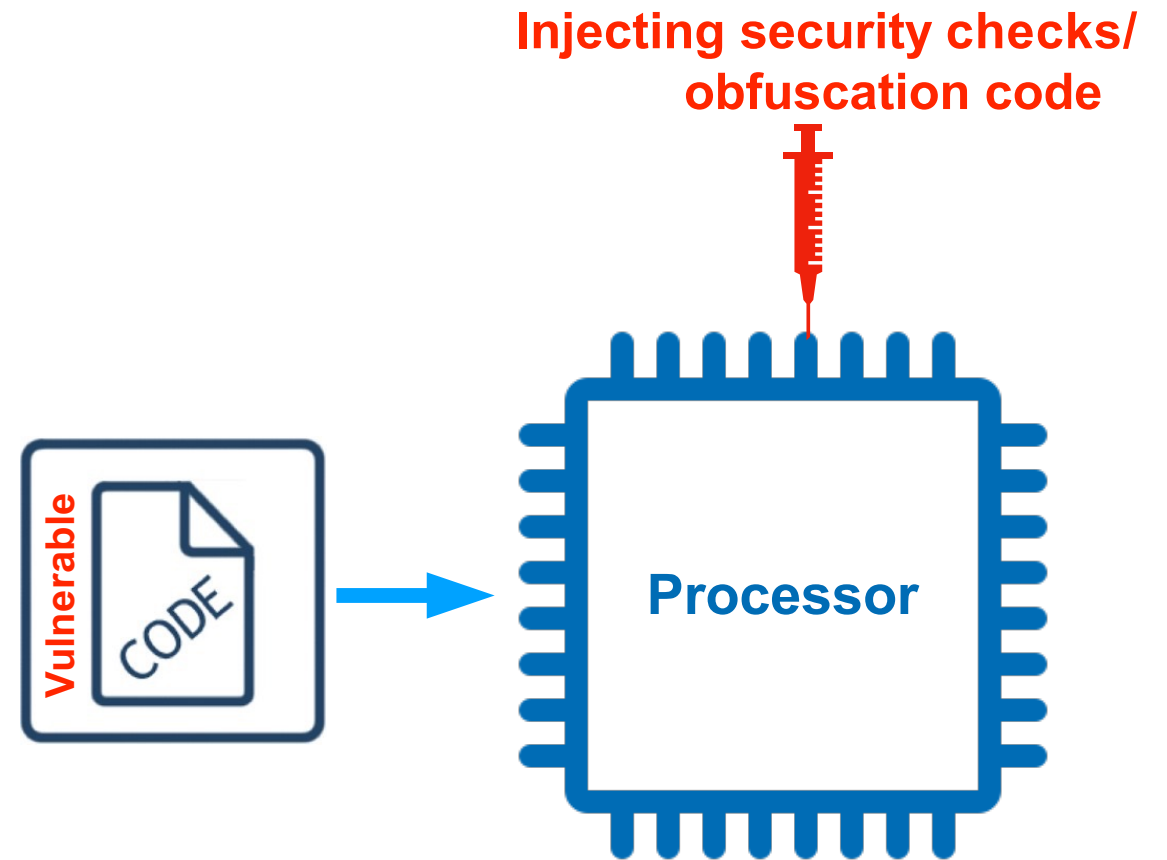
```
void foo ()
{
    ...
    canary_check;
    return;
}
```

```
mov %rcx, 1000000c35841h
mov %rcx, 3BF43B1820E7ED7Dh
mov %rdx, 3BF53B182024B53Ch
xor %rcx, %rdx
mov qword ptr [rax+48h], %rcx
```

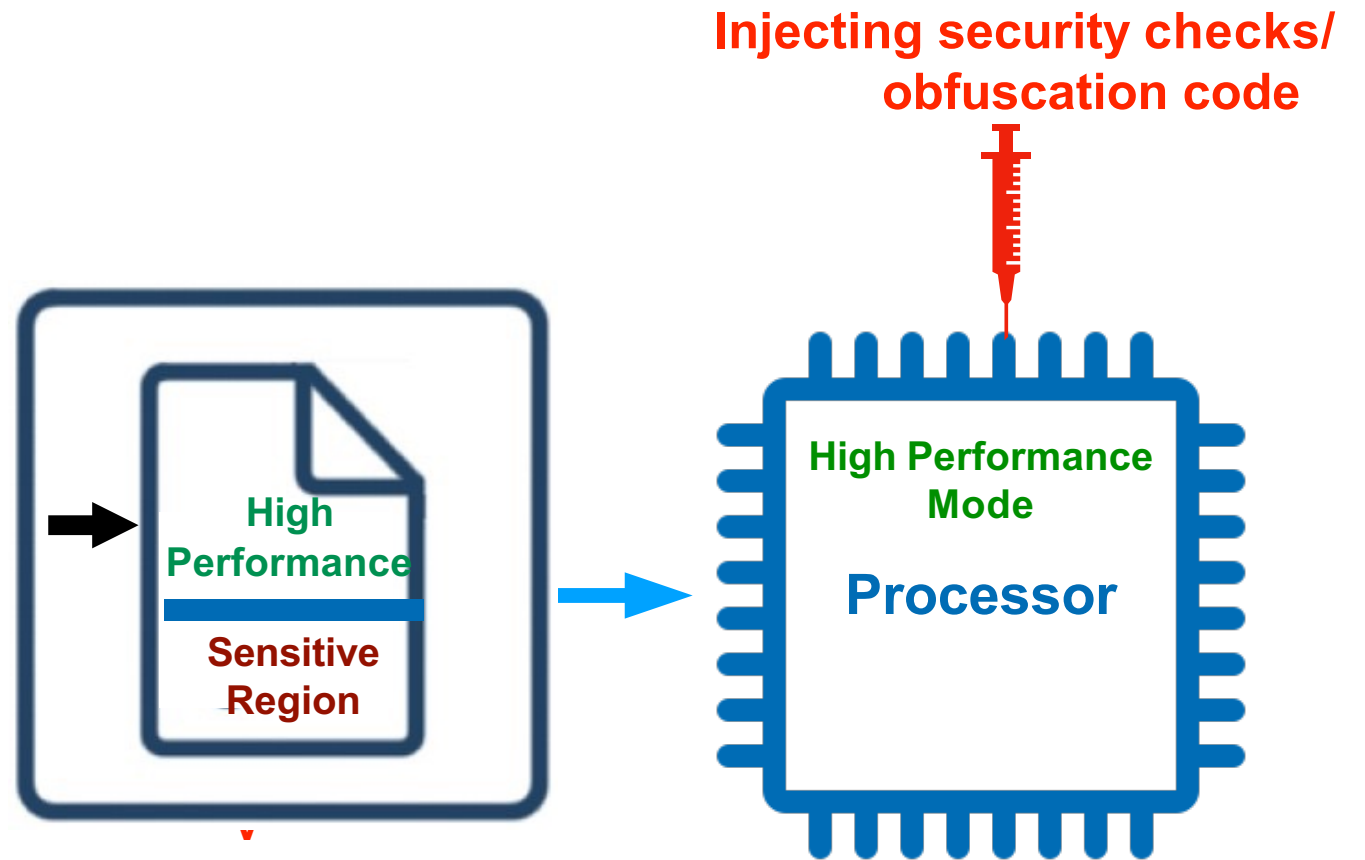
State-of-the-Art Defenses



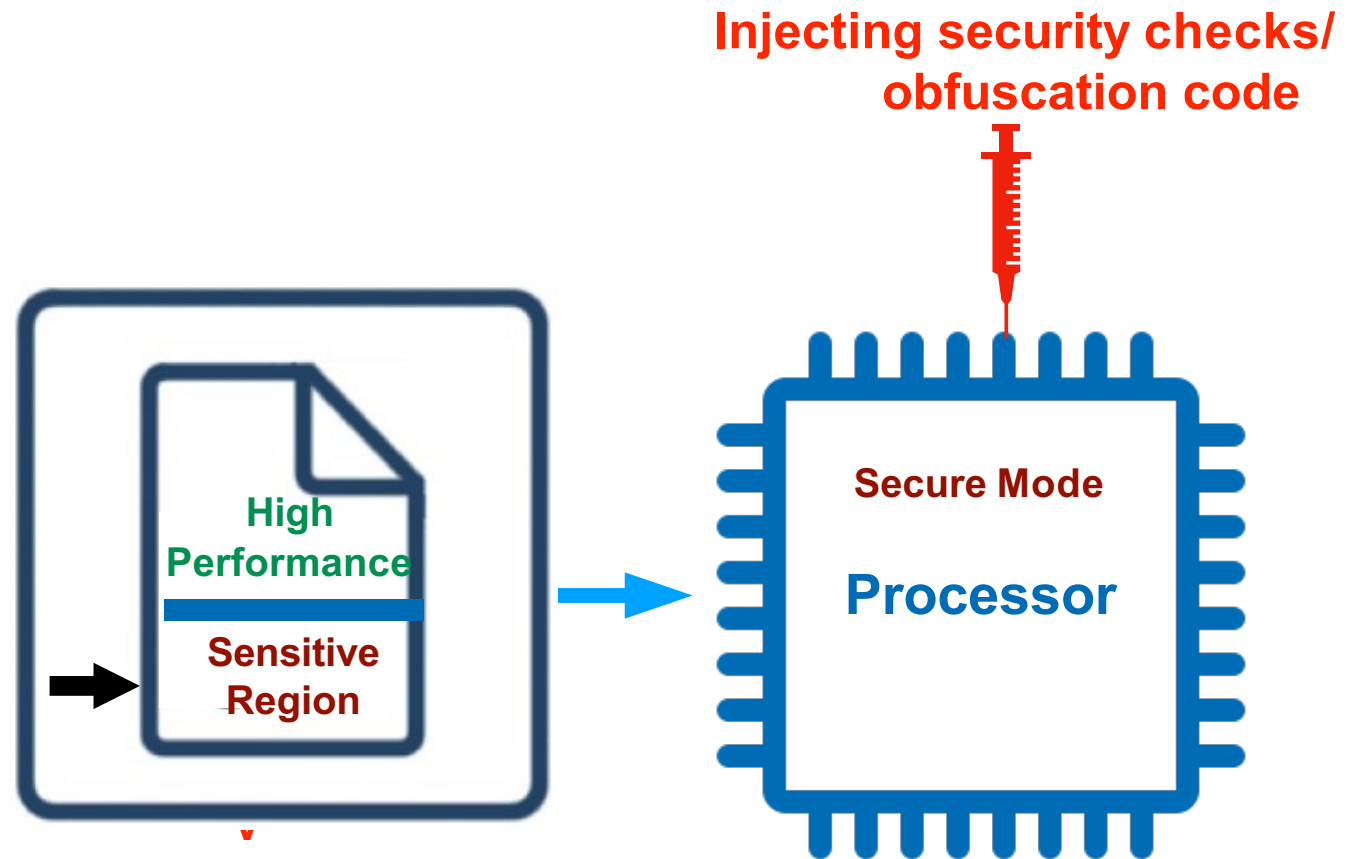
Field-Upgradable Defenses



Field-Upgradable Defenses



Field-Upgradable Defenses



Field-Upgradable Defenses

- ✓ Dynamic
- ✓ Legacy Binaries
- ✓ Cycle-Level Transition Time
- ✓ Invisible to the Attacker

Injecting security checks/
obfuscation code

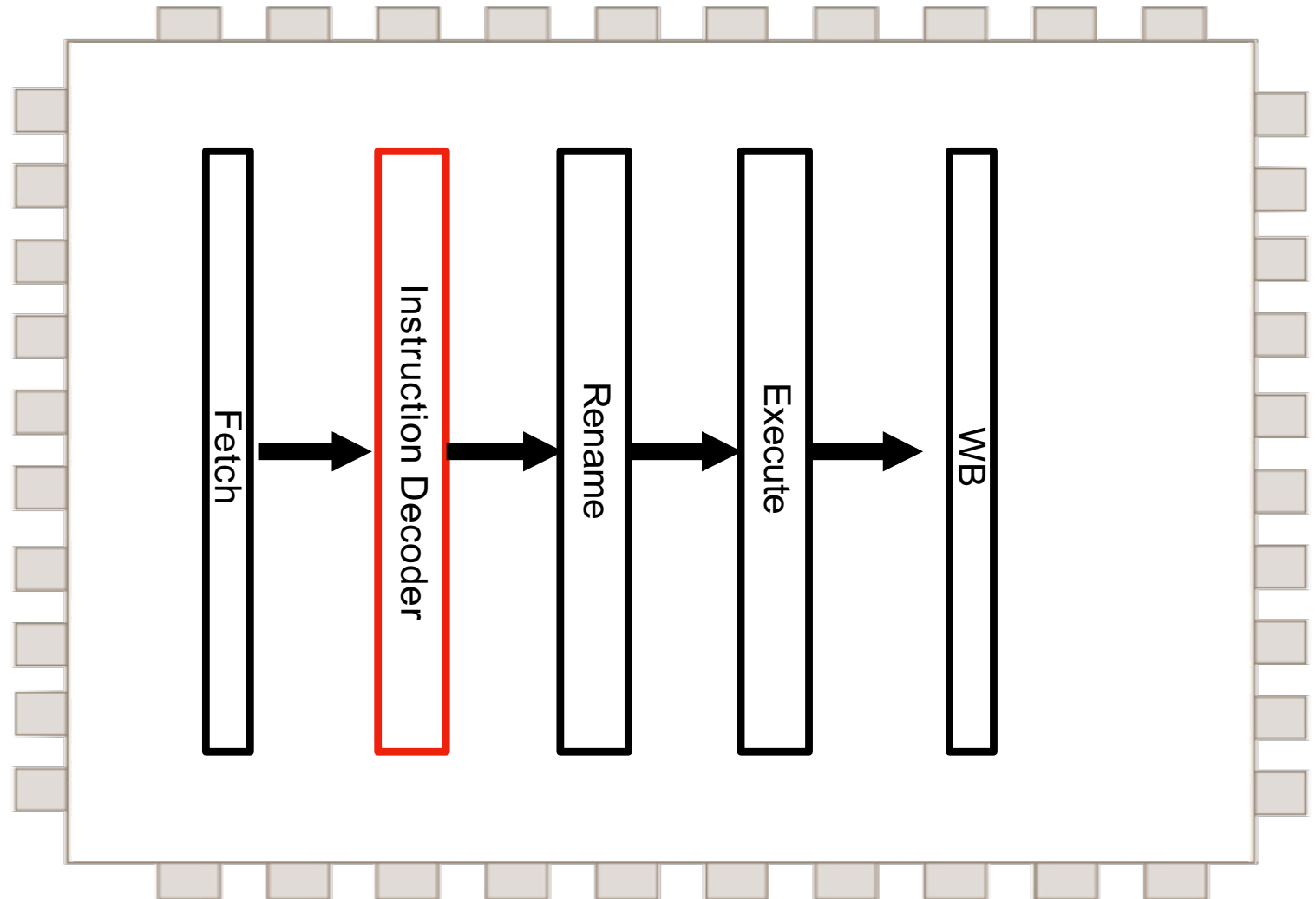


Secure Mode
Processor

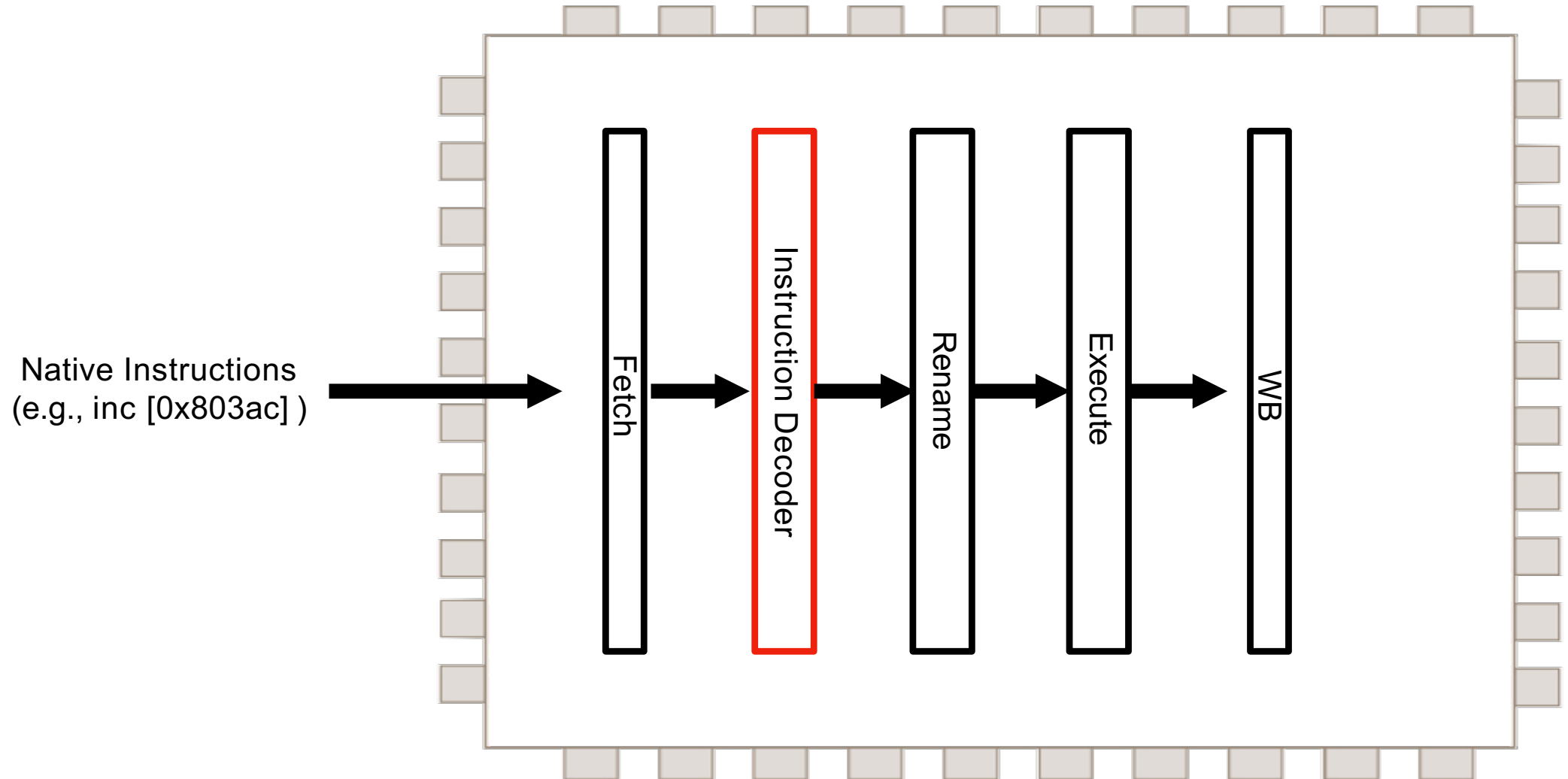
Microcode-based Defenses



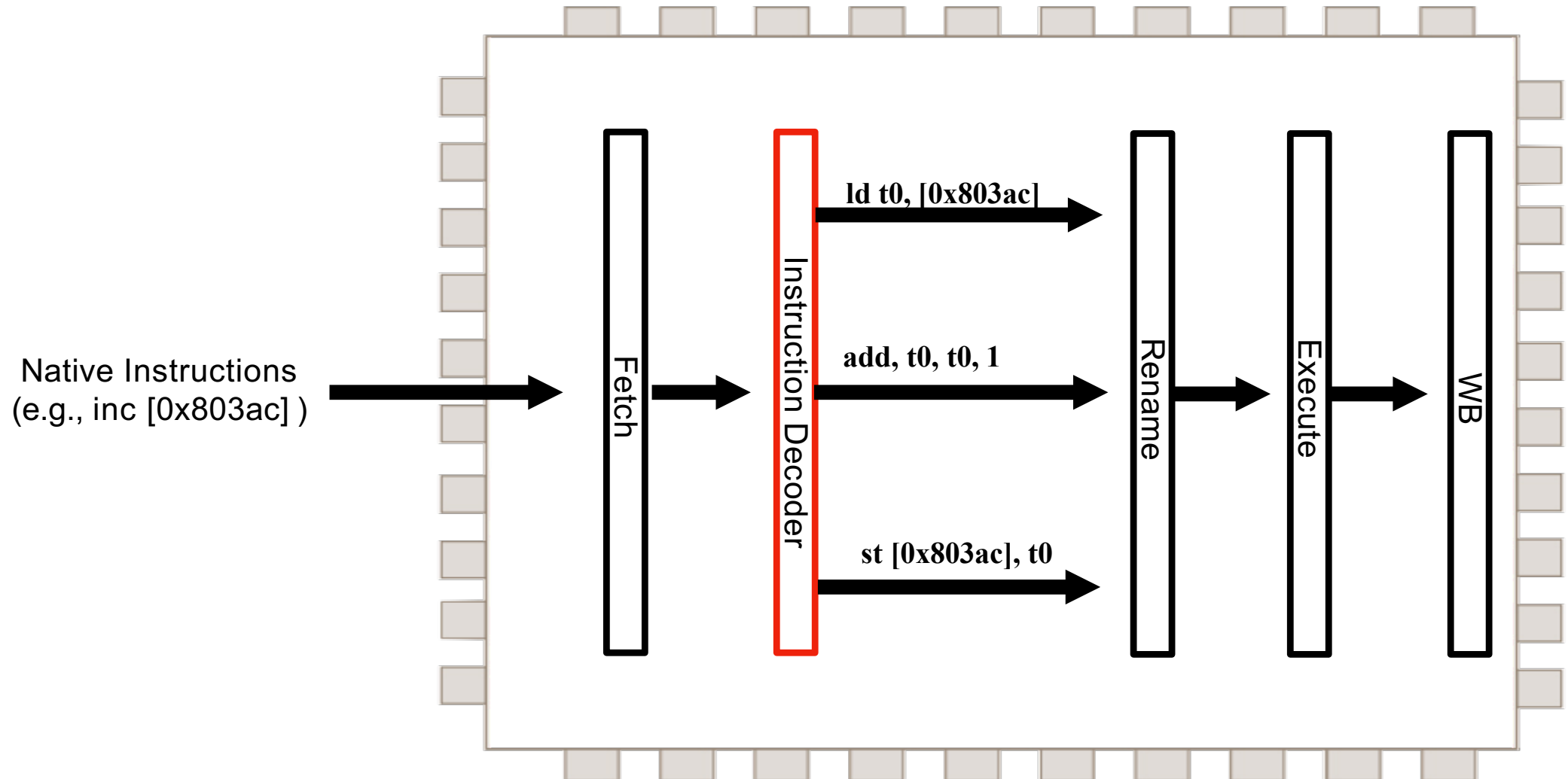
Microcode-based Defenses



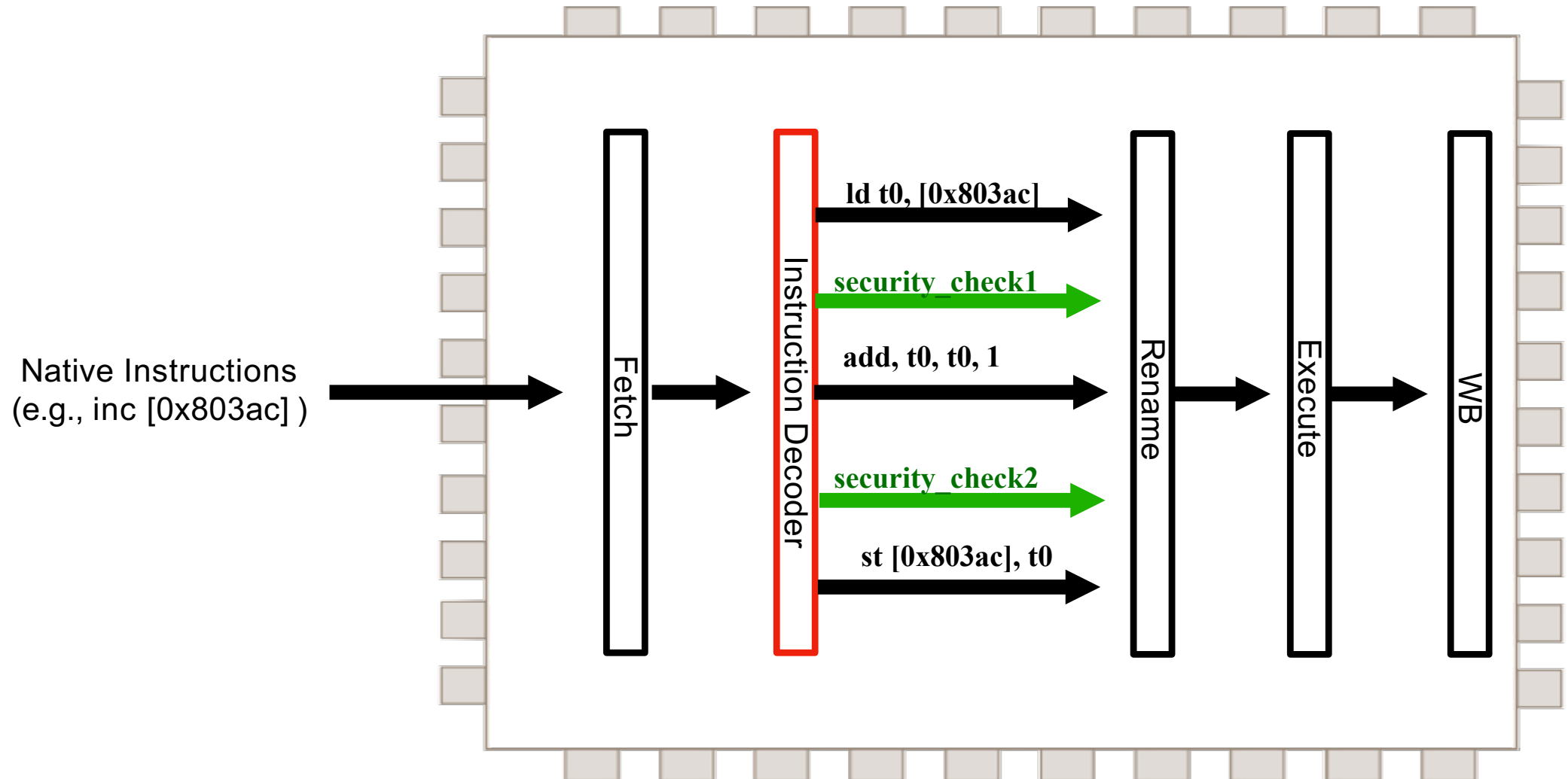
Microcode-based Defenses



Microcode-based Defenses



Microcode-based Defenses





Benefits

- Promotes versatility by enabling software modules to define security properties specific to their domain and language.
- Decouples security specifications from the ISA, allowing dynamic reconfiguration of policies to counter new exploits without recompilation.
- Empowers hardware vendors to adapt enforcement mechanisms transparently with each new generation of hardware.



Challenges

- Strengthening the hardware-software contract is essential for comprehensive defenses against diverse attack vectors.
- Decoupled from the Instruction Set Architecture (ISA) to enable dynamic implementation of security policies.
- Real-time enforcement through innovative microarchitectural mechanisms and runtime monitoring eliminating need for recompilation, redeployment, and frequent hardware upgrades.

Conclusion

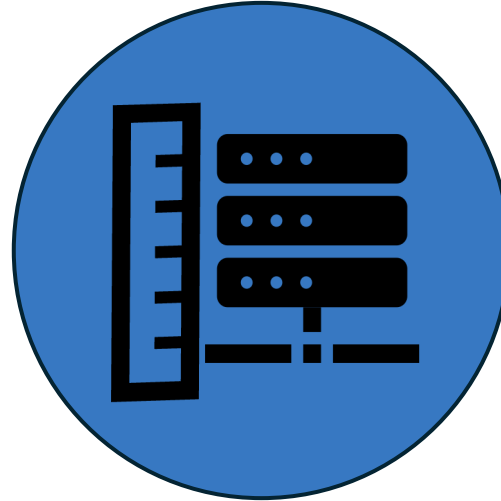
- **Compartmentalization** – isolates compromised hardware components from the rest of the system to prevent the spread of threats.
- **Field-upgradable devices** - reduces costs and decreases the deployment period.
- **Secure virtualization** - eliminates the risks of crosstalk and unauthorized interference by creating isolated virtual environments within the system.

How we have been tackling this!



Transparent and Surgical
Deployment of Security
Enhancements

*(ISCA 2018, IEEE Micro Top Pick
2019, HPCA 2019 Best Paper
Runner-Up, ISCA 2020, IEEE D&T
Hardware Security Top Pick 2020)*



Secure Virtualization of
System Resources

*Ceviche: Capability-Enhanced
Virtualization of Caches,
IEEE S&P 2025*



Compartmentalization

*DARPA Compartmentalization
and Privilege Management
Project*

